

Lab 2

Build your VPC and Launch a Web Server

This lab focuses on accessing the AWS Management Console and creating a Virtual Private Cloud (VPC) with multiple resources in one availability zone, including an Internet Gateway, a public subnet, a private subnet, two route tables, and a NAT Gateway. In Task 1, users will create a VPC with the necessary resources and verify that all the resources are created before proceeding to the next step. Task 2 focuses on creating two additional subnets for the VPC in a second availability zone, providing High Availability for deploying solutions. This lab provides a hands-on approach to learning about AWS VPC and its components.

Reference Video for Lab 2: Build your VPC and Launch a Web Server - (<https://www.youtube.com/watch?v=YT0AccGSJ2k&t=1s>)

This lab is to be completed on AWS Academy Portal, you can find it in under Module 5. You can begin by click **Start Lab** and once the environment is ready click on **AWS**. It will show you ways to access the lab environment. For ease of use there is an option labelled **AWS SSO** which will provide you with a secure url.

Note: Each lab session exists for only 2 hours, and after each session the lab gets destroyed and you have to start over again.

Background

A VPC allows users to create their own virtual network within AWS, providing a level of isolation and security. Web server hosting refers to the process of deploying a website or web application on a server that is accessible over the internet. AWS provides various services to host web servers, including EC2 instances, load balancers, and auto-scaling groups. These services can be configured to provide a highly scalable and fault-tolerant web hosting environment. Additionally, AWS offers various tools and services for monitoring and managing web server instances. With AWS, users can easily set up and manage their web hosting environment, allowing them to focus on developing and delivering their applications to users.

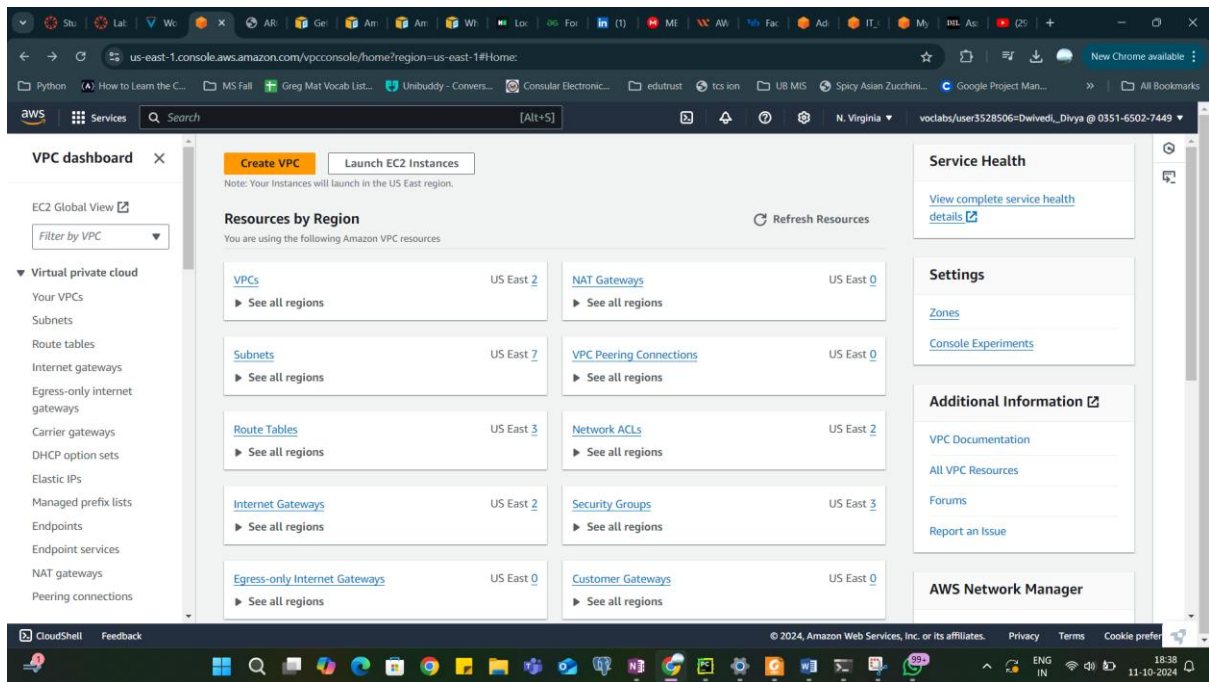
Deliverables

1. A written report that summarizes the key steps and outcomes of the lab exercise, including any challenges faced and lessons learned. This report should also include screenshots of the AWS Management Console and other relevant documentation.

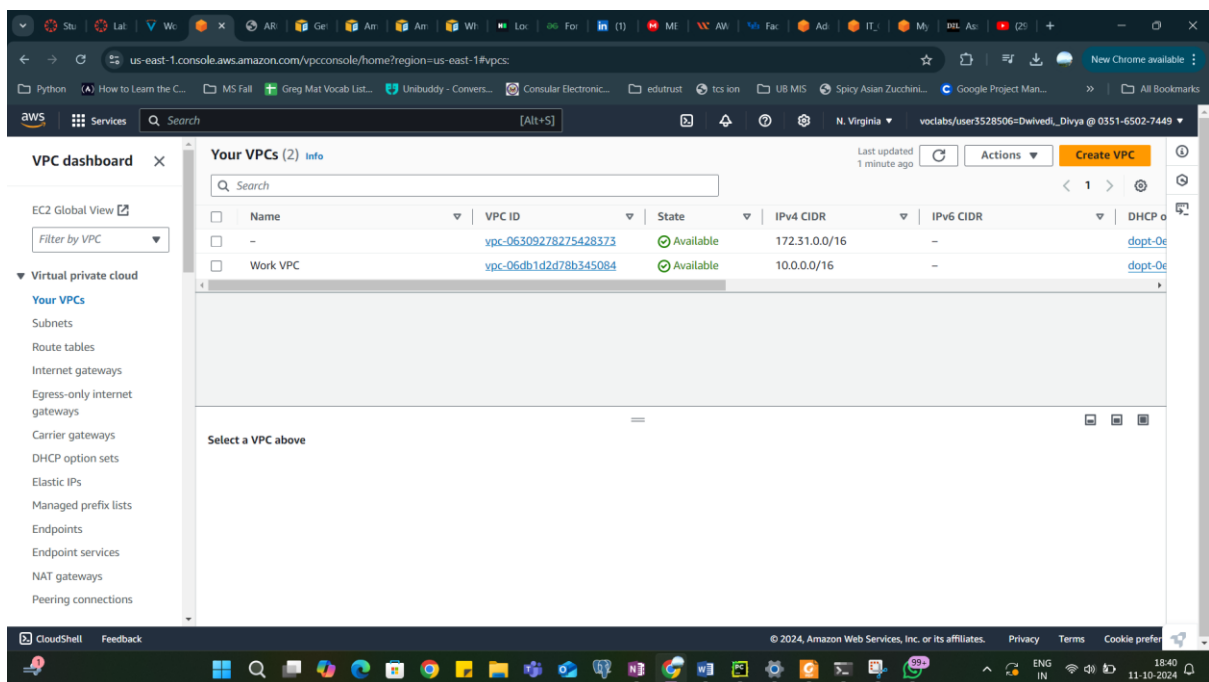
Ans 1) **SCREENSHOTS with steps :**

1. I started by searching VPC upon opening AWS.

MGS 602: Network Technology and Cloud Infrastructure Management

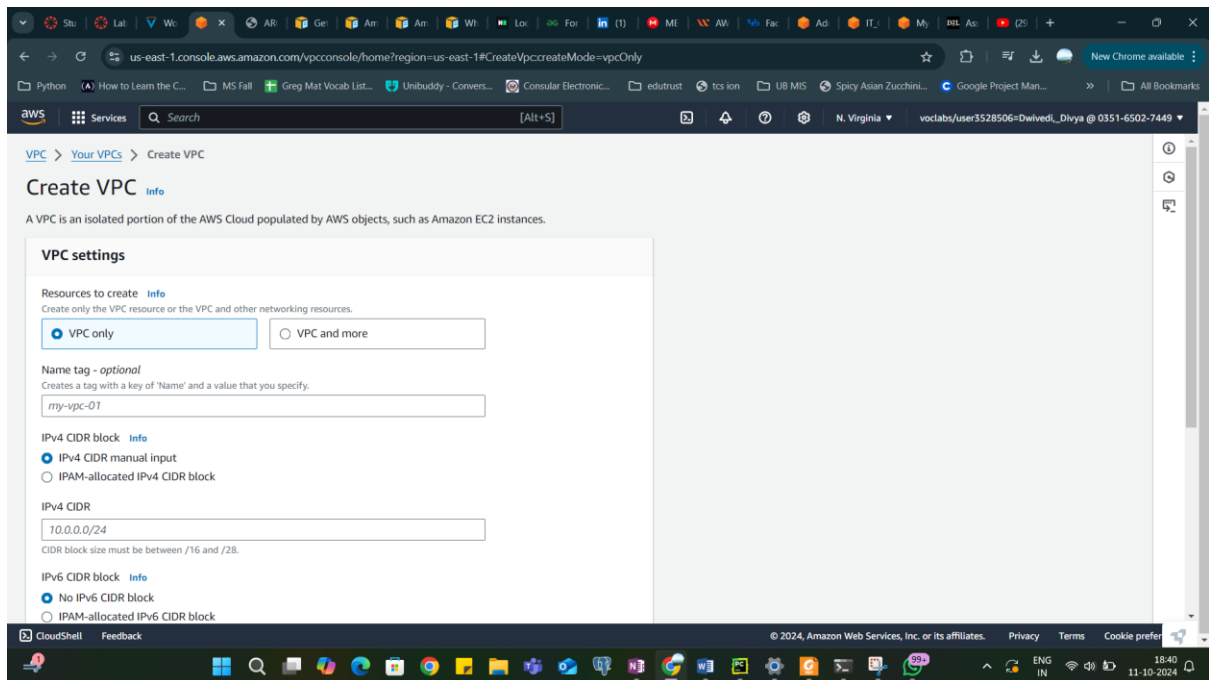


2. Then, I clicked on VPC that appears on the top when you search VPC.

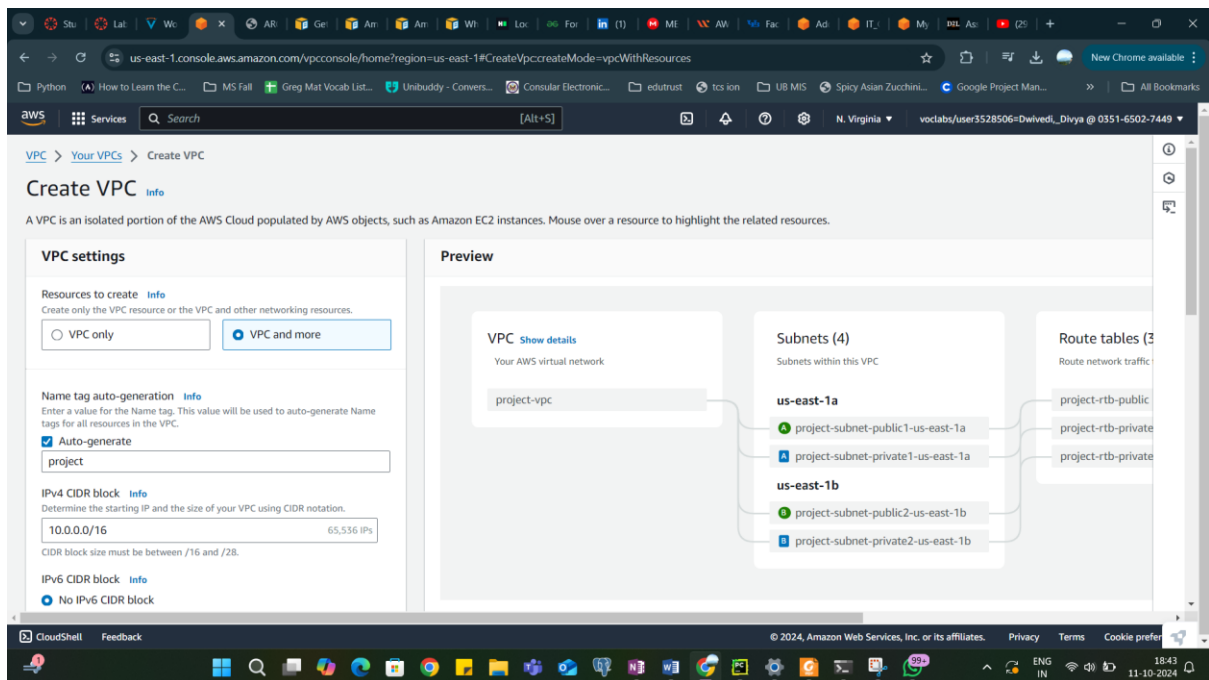


MGS 602: Network Technology and Cloud Infrastructure Management

3. Clicked on Create VPC.

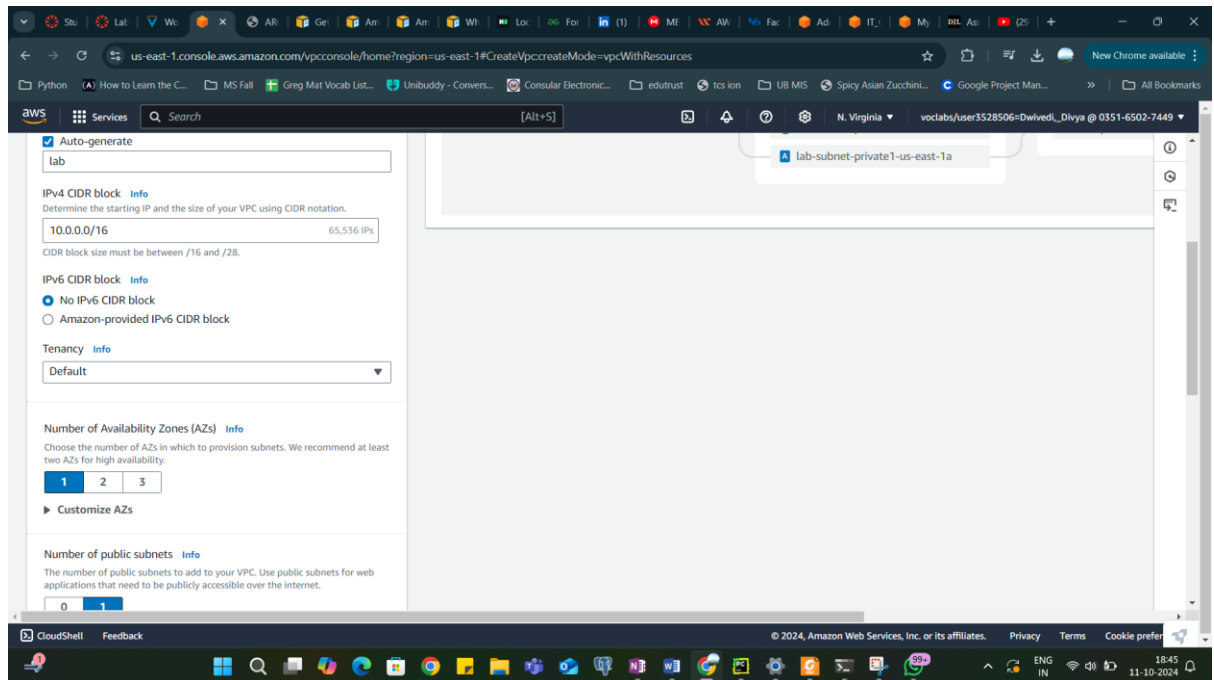


4. Selected VPC and more.

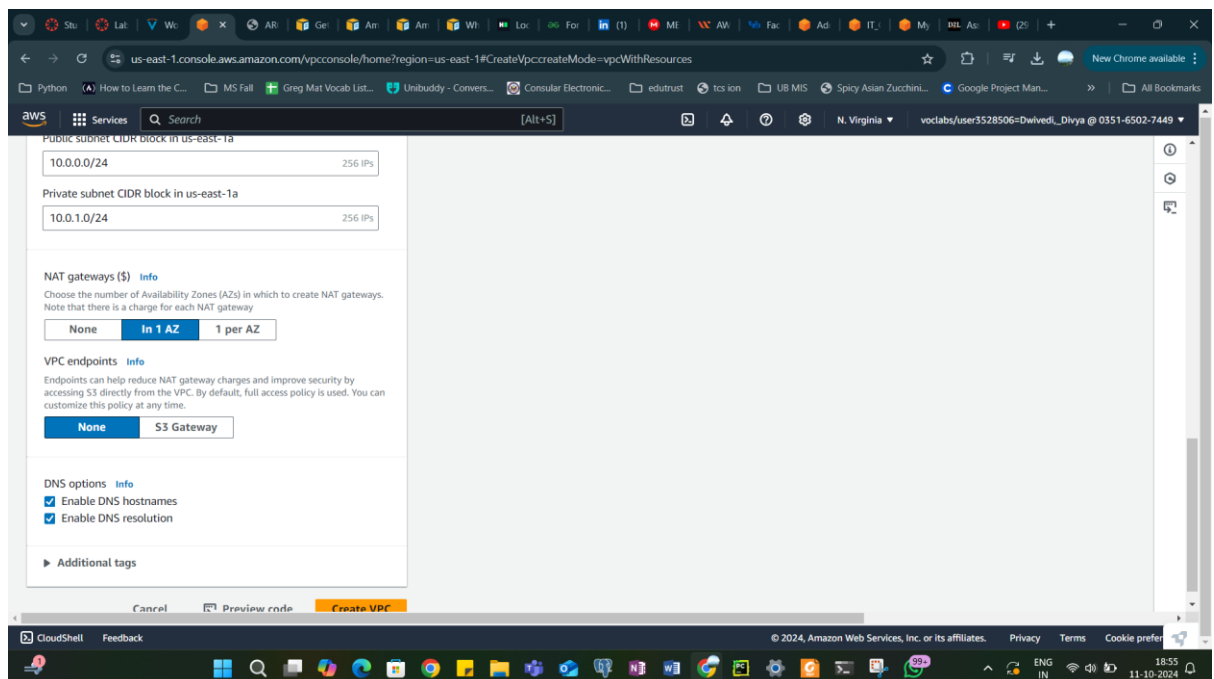


MGS 602: Network Technology and Cloud Infrastructure Management

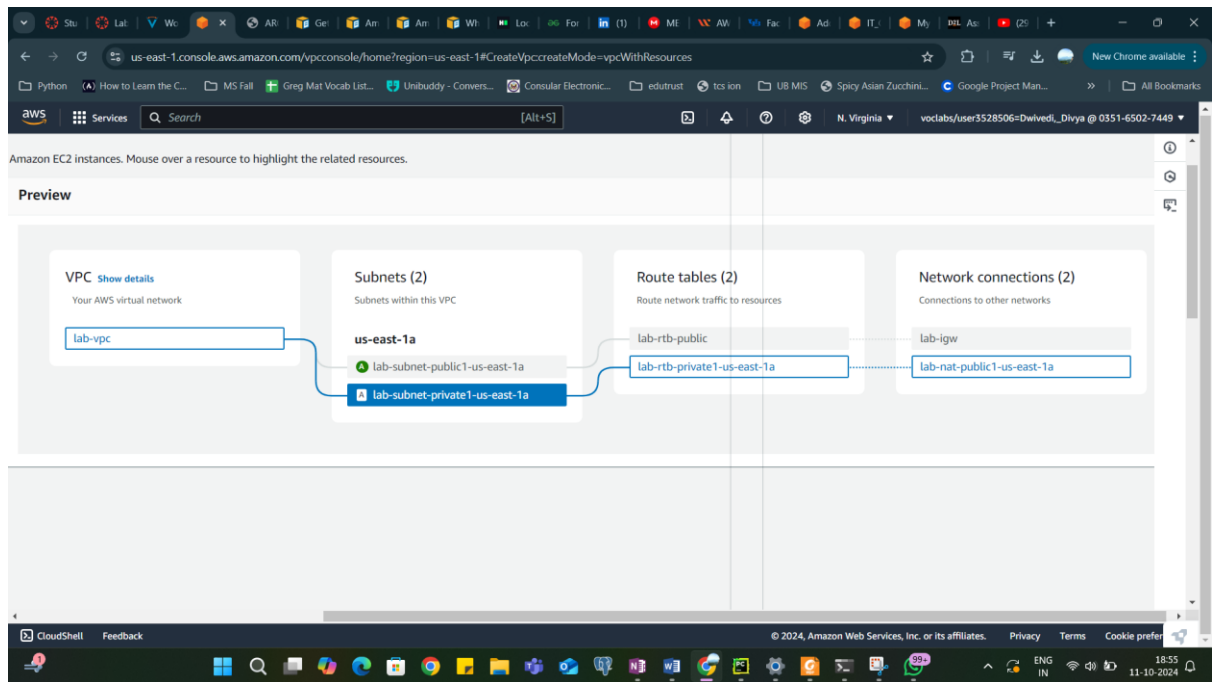
- I set the name tag auto-generation and wrote Lab.
Afterwards, I set the number of Availability Zones (AZs) as 1.



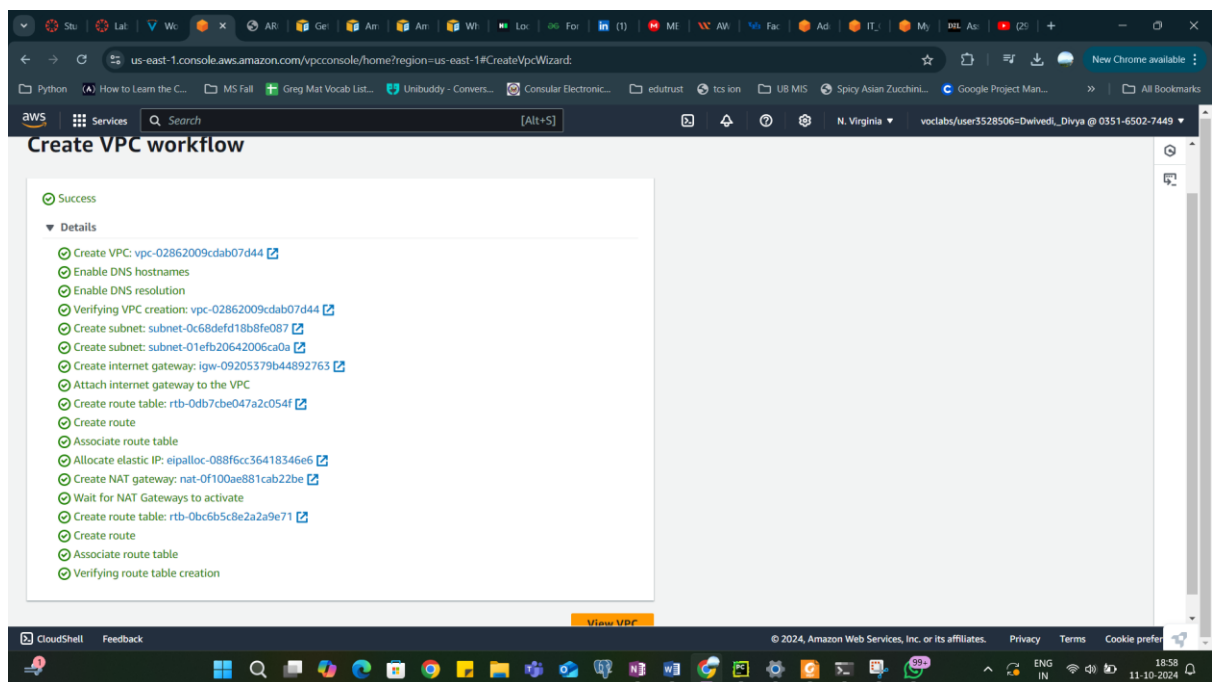
- By clicking on **Customize subnets CIDR blocks**, I changed Public subnet CIDR block in us-east-1a from 10.0.0.0/20 to 10.0.0.0/24, then changed Private subnet CIDR block in us-east-1a from 10.0.128.0/20 to 10.0.1.0/24
Now, I changed NAT Gateways from None to in 1 AZ and I changed VPC endpoints from S3 Gateway to None.



MGS 602: Network Technology and Cloud Infrastructure Management



7. Created Successfully!



MGS 602: Network Technology and Cloud Infrastructure Management

The screenshot shows the AWS VPC console for the VPC `vpc-02862009cdab07d44` in the `us-east-1` region. The console displays the VPC details, including its ID, state (Available), and various configuration options like DHCP, DNS, and route tables. The left sidebar shows the VPC dashboard with options to filter by VPC and view subnets, route tables, internet gateways, etc. The bottom of the console shows a resource map with subnets and route tables.

VPC dashboard **×**

EC2 Global View [↗](#)

Filter by VPC **▼**

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

VPC > **Your VPCs** > vpc-02862009cdab07d44

vpc-02862009cdab07d44 / lab-vpc **Actions** **▼**

Details **Info**

VPC ID vpc-02862009cdab07d44	State Available	DNS hostnames Enabled	DNS resolution Enabled
Tenancy Default	DHCP option set dopt-0e721dbd354daa89f	Main route table rtb-05ac7a21e9e04f0a6	Main network ACL acl-09bc3d15ee525b6da
Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -	IPv6 CIDR (Network border group) -
Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 035165027449	

Resource map **Info**

Resource map **Info**

VPC [Show details](#)

Subnets (2)

Route tables (3)

Net

CloudShell [Feedback](#)

© 2024, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

ENG IN 18:59 11-10-2024

8. Created Additional Subnets.

The screenshot shows the AWS VPC console for the VPC `vpc-06309278275428373` in the `us-east-1` region. The console displays a list of subnets, including their names, IDs, states, VPCs, and IPv4 CIDR blocks. The left sidebar shows the VPC dashboard with options to filter by VPC and view subnets, route tables, internet gateways, etc. The bottom of the console shows a resource map with subnets and route tables.

VPC dashboard **×**

EC2 Global View [↗](#)

Filter by VPC **▼**

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

VPC > **Your VPCs** > vpc-06309278275428373

Subnets (7) **Info**

Find resources by attribute or tag

Last updated 22 minutes ago **Actions** **▼** **Create subnet**

☐	Name	Subnet ID	State	VPC	IPv4 CIDR
☐	-	subnet-0a6fd629c96dd3f20	Available	vpc-06309278275428373	172.31.64.0/20
☐	Work Public Subnet	subnet-0927c14fdef299086	Available	vpc-06db1d2d78b345084 Wo...	10.0.0.0/24
☐	-	subnet-09591dfdb6c25453c	Available	vpc-06309278275428373	172.31.48.0/20
☐	-	subnet-0d00ba59f0503e93	Available	vpc-06309278275428373	172.31.0.0/20
☐	-	subnet-072a77e172c4c93e5	Available	vpc-06309278275428373	172.31.32.0/20
☐	-	subnet-0c42563490ec1cb75	Available	vpc-06309278275428373	172.31.80.0/20
☐	-	subnet-0d377875hd9a5335c	Available	vpc-06309278275428373	172.31.16.0/20

Select a subnet

CloudShell [Feedback](#)

© 2024, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

ENG IN 19:00 11-10-2024

MGS 602: Network Technology and Cloud Infrastructure Management

The screenshot shows the AWS Management Console interface for creating a new subnet. The browser address bar indicates the URL: `us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSubnet`. The page title is "CreateSubnet".

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs
Navigation arrows: < > ^ v

Tags - optional

Key	Value - optional	
Name	lab-subnet-public2	Remove

You can add 49 more tags.

© 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

The screenshot shows the AWS Management Console interface for creating a new subnet. The browser address bar indicates the URL: `us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSubnet`. The page title is "CreateSubnet".

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs
Navigation arrows: < > ^ v

Tags - optional

Key	Value - optional	
Name	lab-subnet-private2	Remove

You can add 49 more tags.

© 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

MGS 602: Network Technology and Cloud Infrastructure Management

9. Then I clicked on Route tables, subnet associations and then edit subnet associations.

The screenshot shows the AWS VPC console interface. On the left, the 'VPC dashboard' sidebar is visible with various navigation options. The main content area displays 'Route tables (1/6)'. A table lists several route tables, with 'lab-rtb-private1-us-east-1a' selected. Below the table, the 'Routes' tab is active, showing a list of two routes. The first route has a destination of 0.0.0.0/0 and a target of nat-0f100ae881cab22be, with an 'Active' status. The second route has a destination of 10.0.0.0/16 and a target of local, also with an 'Active' status.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
lab-rtb-public	rtb-0db7cbe047a2c054f	subnet-0c68defd18b8fe0...	-	No	vpc-02862009cdab07d44
Work Public Route Table	rtb-0fbfe05fd335d047	subnet-0927c14def2990...	-	No	vpc-06db1d2d78b345084
-	rtb-0c9f089989771c6	-	-	Yes	vpc-06db1d2d78b345084
-	rtb-05ac7a21e9e04f0a6	-	-	Yes	vpc-02862009cdab07d44
-	rtb-0a2ae7e0e1e1b14d1	-	-	Yes	vpc-06309278275428373
lab-rtb-private1-us-east-1a	rtb-0bc6b5c8e2a2a9e71	subnet-01efb20642006c...	-	No	vpc-02862009cdab07d44

Destination	Target	Status	Propagated
0.0.0.0/0	natur-0f100ae881cab22be	Active	No
10.0.0.0/16	local	Active	No

10. Selected private2 as shown in the screenshot and clicked on Save configurations.

The screenshot shows the 'Edit subnet associations' page in the AWS VPC console. The page title is 'Edit subnet associations' with a subtitle 'Change which subnets are associated with this route table.' Below this, there are two sections: 'Available subnets (2/4)' and 'Selected subnets'. The 'Available subnets' section contains a table with four subnets. The 'lab-subnet-private2' subnet is selected. The 'Selected subnets' section shows the selected subnets: 'subnet-01efb20642006ca0a / lab-subnet-private1-us-east-1a' and 'subnet-0c44379322980373f / lab-subnet-private2'. At the bottom right, there are 'Cancel' and 'Save associations' buttons.

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
lab-subnet-public1-us-east-1a	subnet-0c68defd18b8fe087	10.0.0.0/24	-	rtb-0db7cbe047a2c054f / lab-rtb-public
lab-subnet-private1-us-east-1a	subnet-01efb20642006ca0a	10.0.1.0/24	-	rtb-0bc6b5c8e2a2a9e71 / lab-rtb-priv...
lab-subnet-public2	subnet-0d37c3d995f7304ed	10.0.2.0/24	-	Main (rtb-05ac7a21e9e04f0a6)
lab-subnet-private2	subnet-0c44379322980373f	10.0.3.0/24	-	Main (rtb-05ac7a21e9e04f0a6)

Selected subnets

subnet-01efb20642006ca0a / lab-subnet-private1-us-east-1a X subnet-0c44379322980373f / lab-subnet-private2 X

Cancel Save associations

MGS 602: Network Technology and Cloud Infrastructure Management

11. Clicked upon public and edited subnet associations.

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- Endpoints
- Endpoint services
- NAT gateways
- Peering connections

Route tables (1/6) info

Find resources by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒ lab-rtb-public	rtb-0db7cbe047a2c054f	subnet-0c68defd18b8fe0...	-	No	vpc-02862009cdab07d...
☐ Work Public Route Table	rtb-0fbfe05fd3535d047	subnet-0927c14fdef2990...	-	No	vpc-06db1d2d78b3450...
☐ -	rtb-0c19f0899989771c6	-	-	Yes	vpc-06db1d2d78b3450...
☐ -	rtb-05ac7a21e9e04f0a6	-	-	Yes	vpc-02862009cdab07d...
☐ -	rtb-0a2ae7e0e1e1b14d1	-	-	Yes	vpc-063092782754283...
☐ lab-rtb-private1-us-east-1a	rtb-0bc6b5c8e2a2a9e71	2 subnets	-	No	vpc-02862009cdab07d...

rtb-0db7cbe047a2c054f / lab-rtb-public

Details | Routes | **Subnet associations** | Edge associations | Route propagation | Tags

Explicit subnet associations (1)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
lab-subnet-public1-us-east-1a	subnet-0c68defd18b8fe087	10.0.0.0/24	-

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒ lab-subnet-public1-us-east-1a	subnet-0c68defd18b8fe087	10.0.0.0/24	-	rtb-0db7cbe047a2c054f / lab-rtb-public
☐ lab-subnet-private1-us-east-1a	subnet-01efb20642006ca0a	10.0.1.0/24	-	rtb-0bc6b5c8e2a2a9e71 / lab-rtb-priv...
☒ lab-subnet-public2	subnet-0d37c3d995f7304ed	10.0.2.0/24	-	Main (rtb-05ac7a21e9e04f0a6)
☐ lab-subnet-private2	subnet-0c44379322980373f	10.0.3.0/24	-	rtb-0bc6b5c8e2a2a9e71 / lab-rtb-priv...

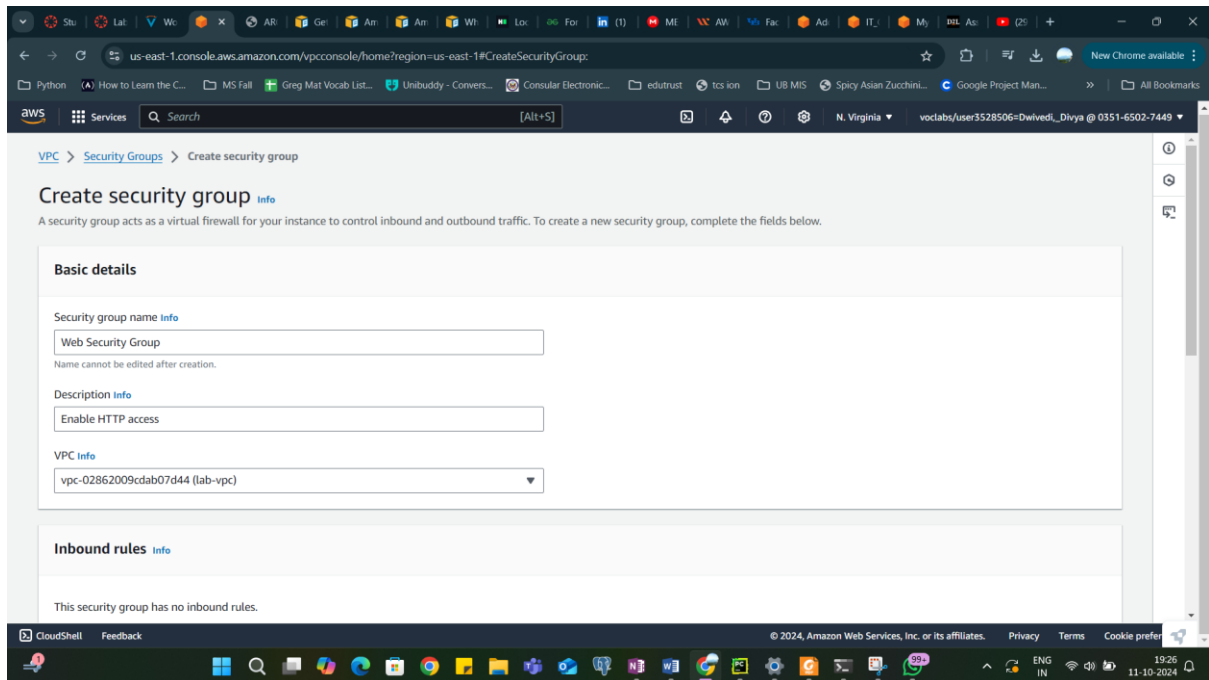
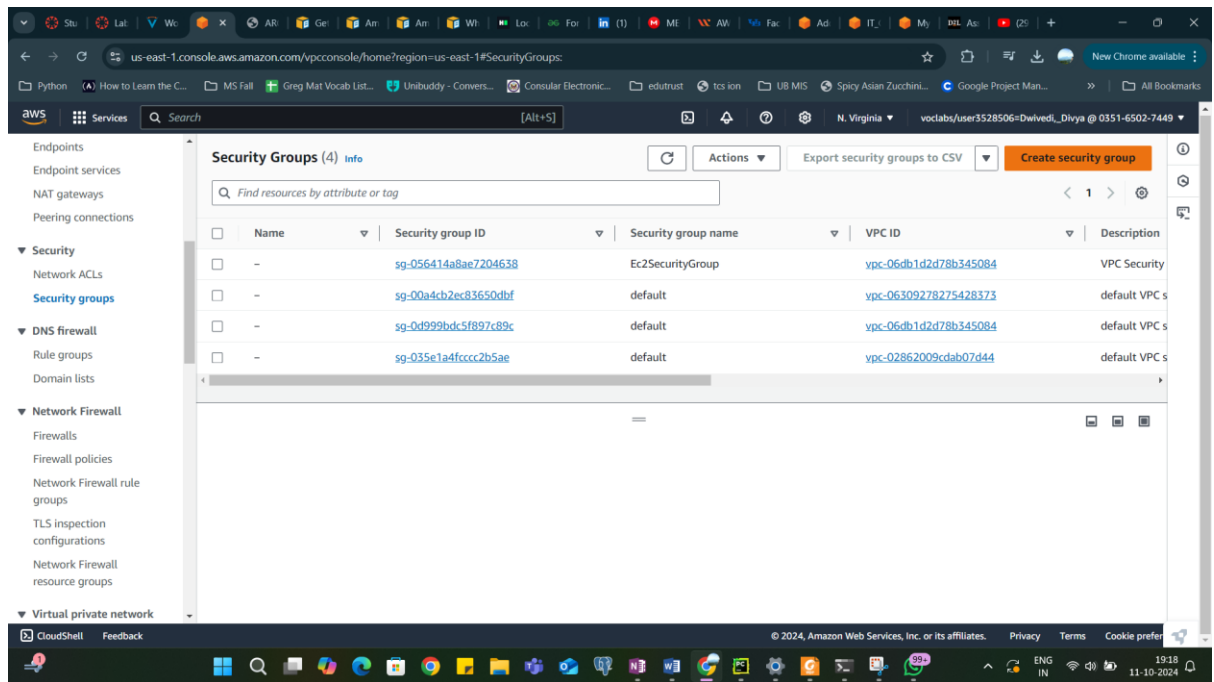
Selected subnets

subnet-0c68defd18b8fe087 / lab-subnet-public1-us-east-1a X subnet-0d37c3d995f7304ed / lab-subnet-public2 X

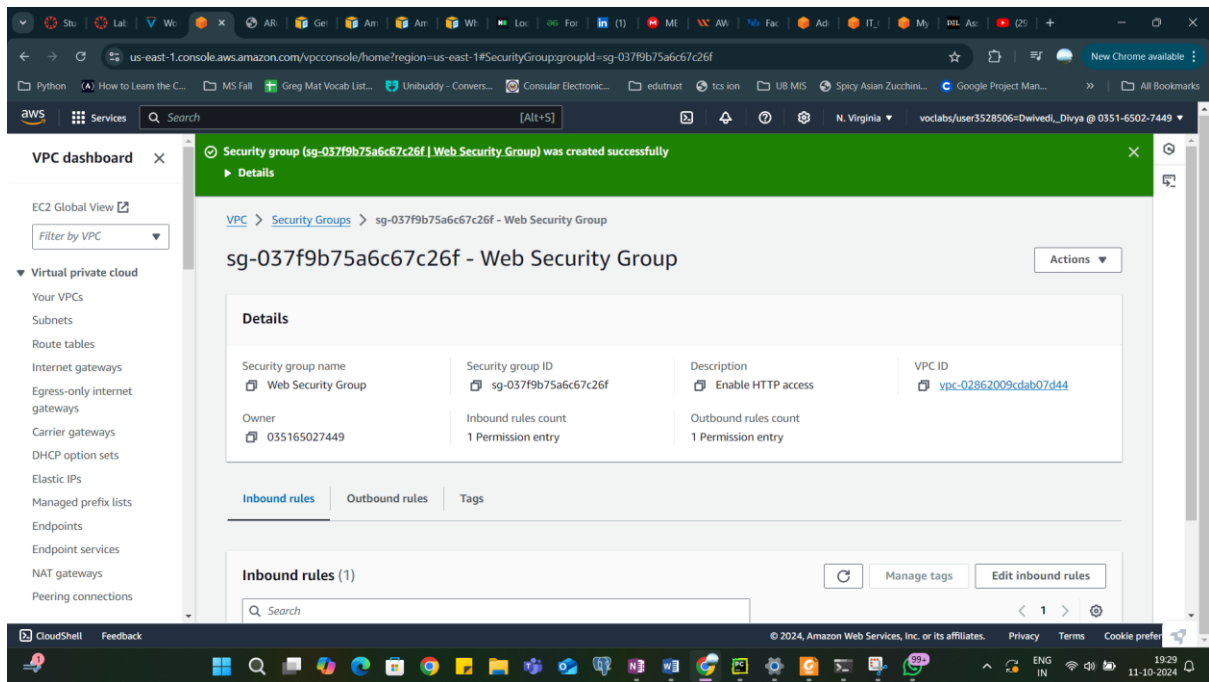
Cancel Save associations

MGS 602: Network Technology and Cloud Infrastructure Management

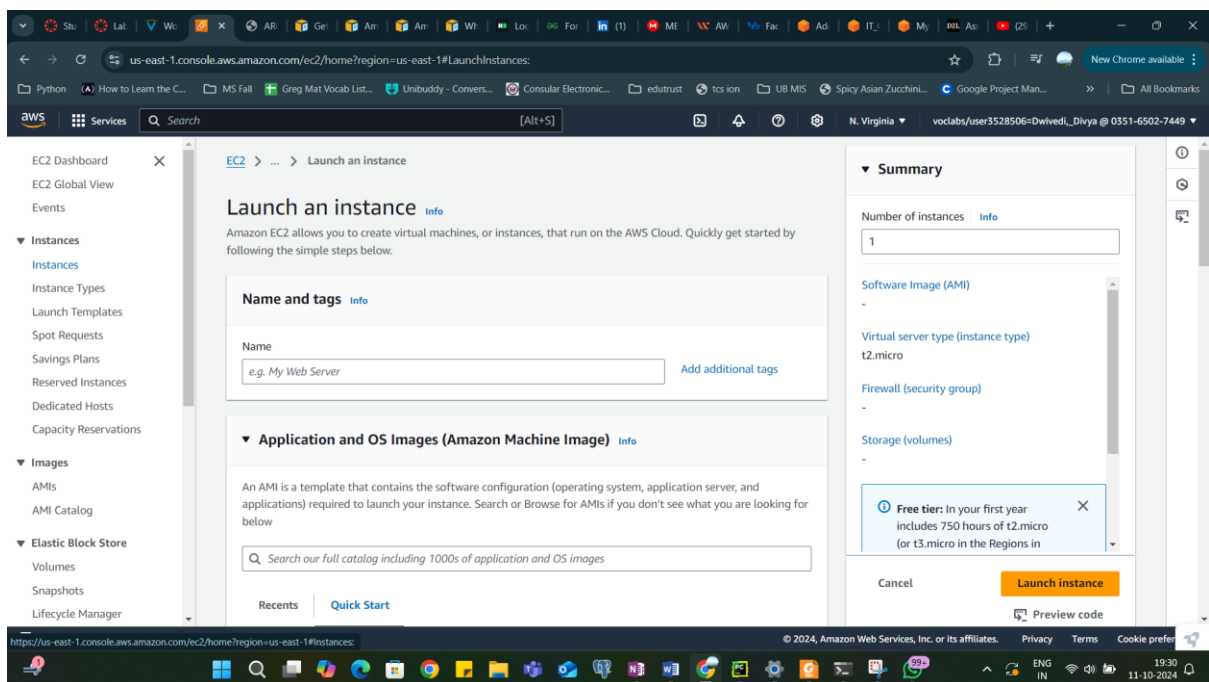
12. Then I created a Security Group by adding inbound rules.



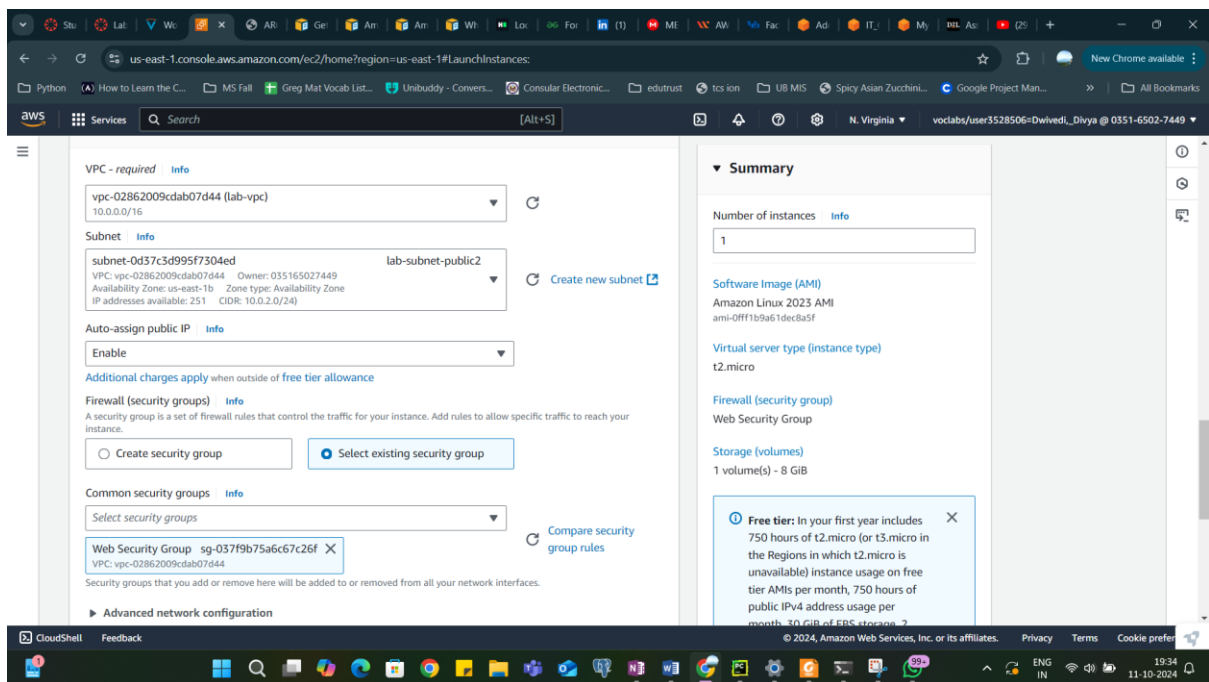
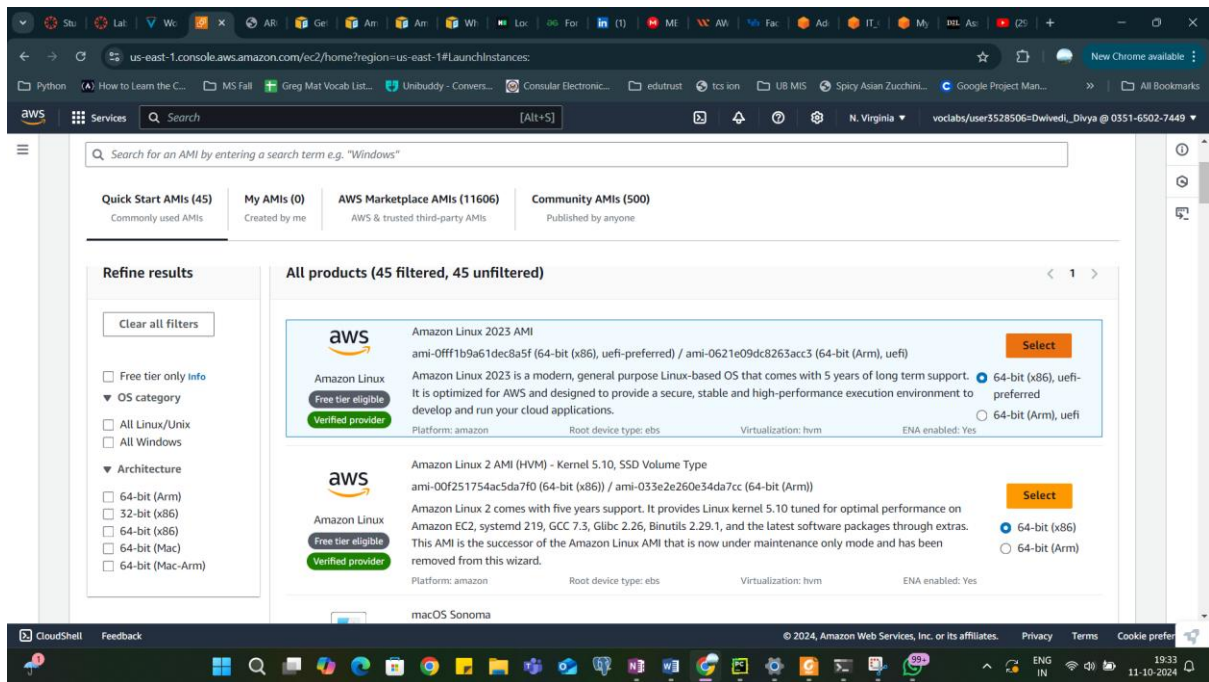
MGS 602: Network Technology and Cloud Infrastructure Management



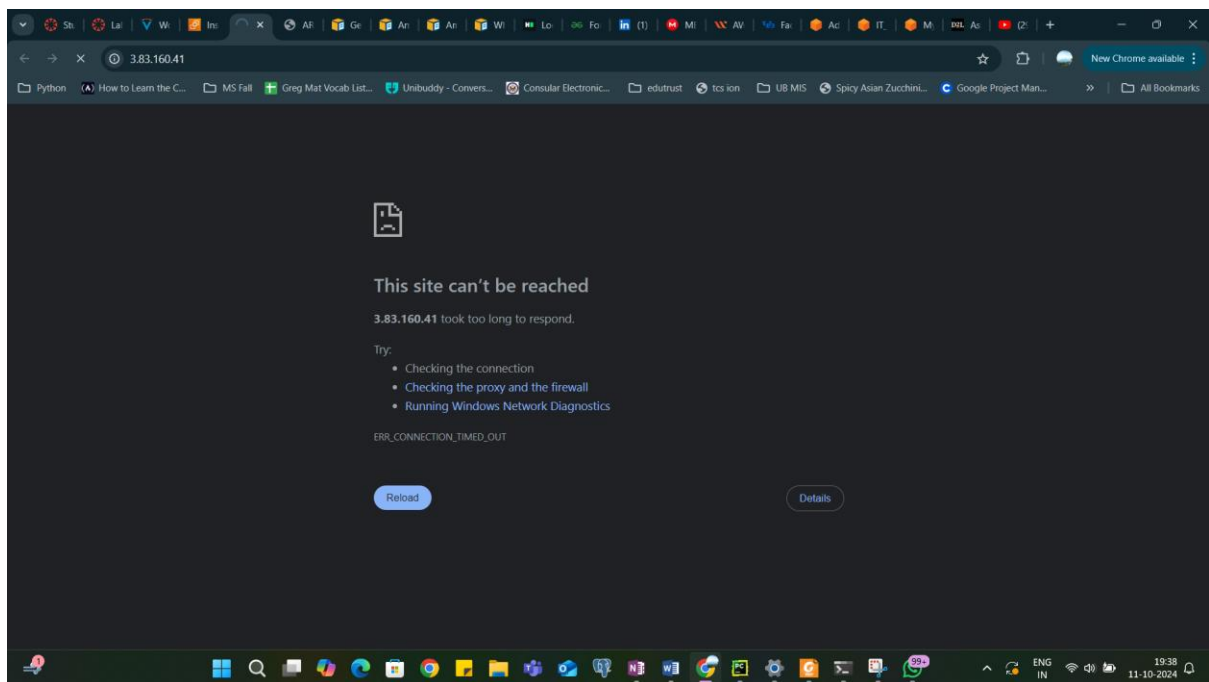
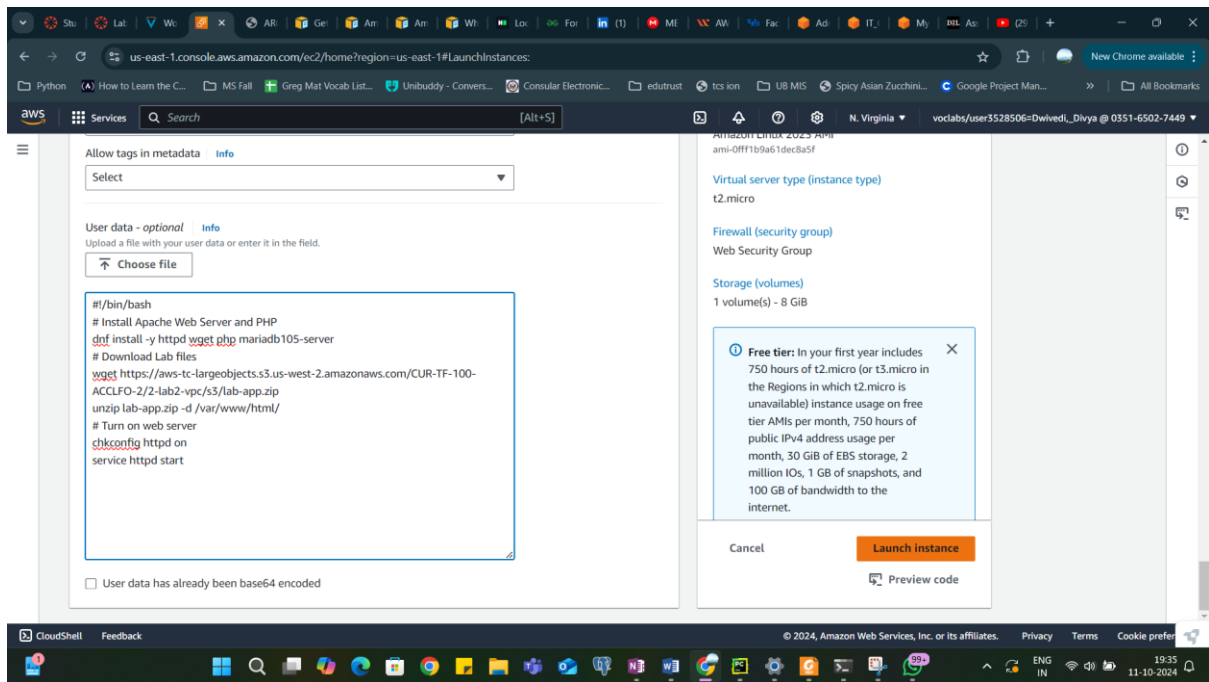
13. Searched Ec2, clicked on Launch Instance and then Instances from the side bar.



MGS 602: Network Technology and Cloud Infrastructure Management

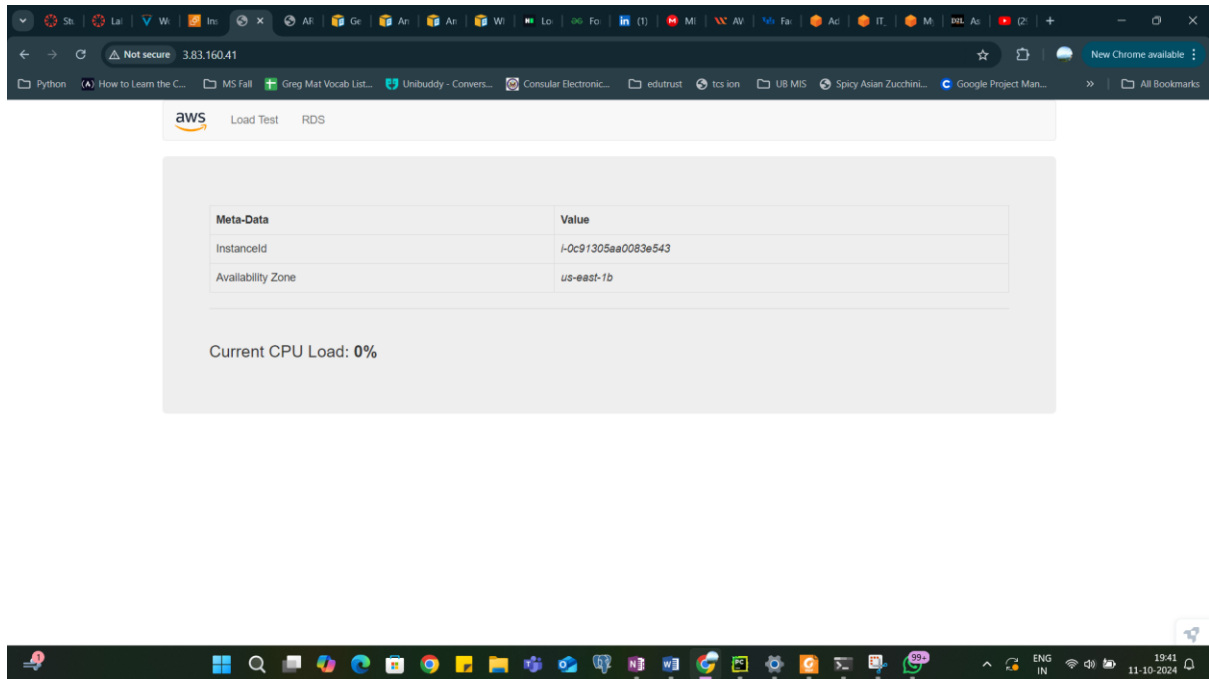


MGS 602: Network Technology and Cloud Infrastructure Management



MGS 602: Network Technology and Cloud Infrastructure Management

14. Lab ran successfully! IPPv4 address is now loaded.

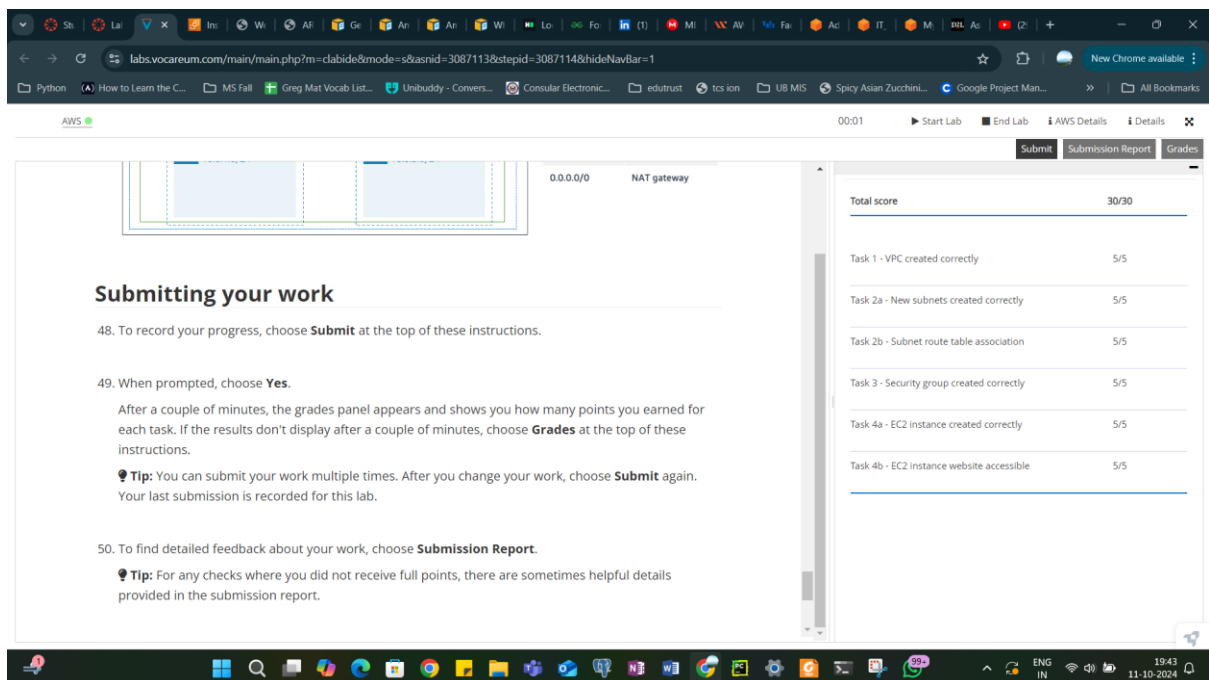


The screenshot shows the AWS Load Test console. At the top, there's a navigation bar with 'aws', 'Load Test', and 'RDS'. Below this, a table displays instance metadata:

Meta-Data	Value
InstanceId	i-0c91305aa0083e543
Availability Zone	us-east-1b

Below the table, it states 'Current CPU Load: 0%'. The browser's address bar shows '3.83.160.41'.

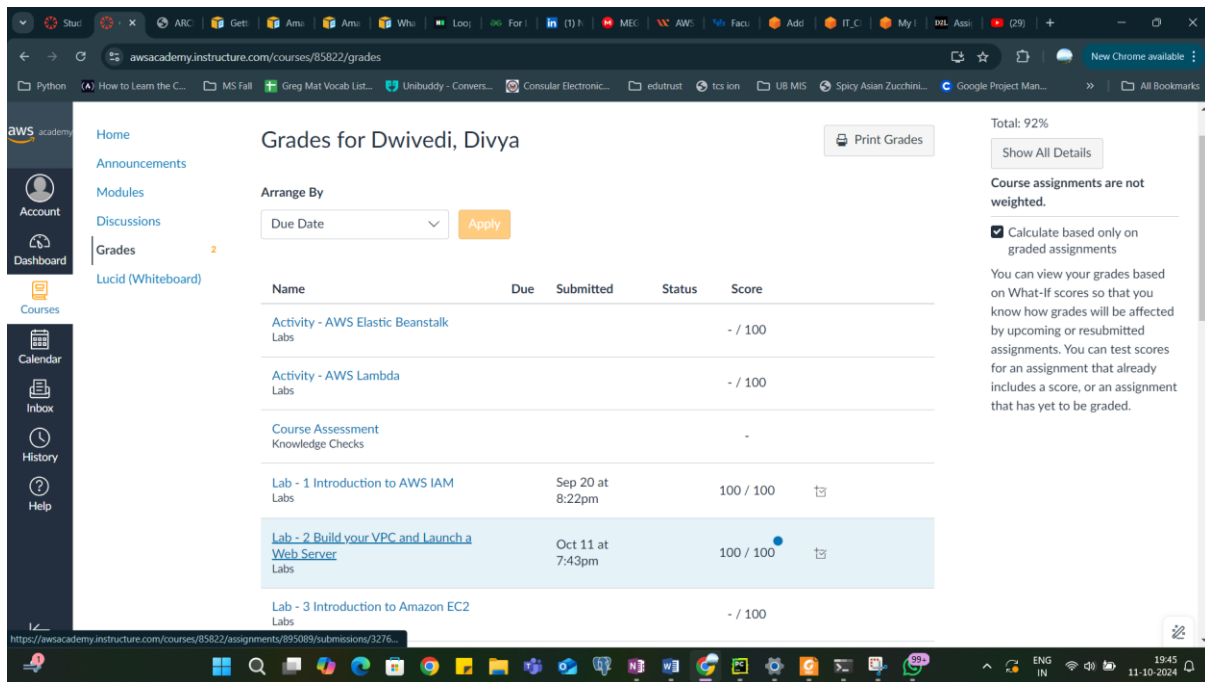
15. Lab submitted successfully!



The screenshot shows the lab submission interface. On the left, there's a section titled 'Submitting your work' with instructions 48, 49, and 50. Instruction 48 says to choose 'Submit' at the top. Instruction 49 says to choose 'Yes' and mentions that the grades panel will appear after a few minutes. Instruction 50 says to choose 'Submission Report' for detailed feedback. On the right, there's a 'Grades' panel showing the total score and individual task scores:

Task	Score
Total score	30/30
Task 1 - VPC created correctly	5/5
Task 2a - New subnets created correctly	5/5
Task 2b - Subnet route table association	5/5
Task 3 - Security group created correctly	5/5
Task 4a - EC2 instance created correctly	5/5
Task 4b - EC2 instance website accessible	5/5

MGS 602: Network Technology and Cloud Infrastructure Management



The screenshot shows the AWS Academy Instructure interface for a user named Dwivedi, Divya. The page displays a list of activities and labs with their respective scores. The total score is 92%.

Name	Due	Submitted	Status	Score
Activity - AWS Elastic Beanstalk Labs				- / 100
Activity - AWS Lambda Labs				- / 100
Course Assessment Knowledge Checks				-
Lab - 1 Introduction to AWS IAM Labs	Sep 20 at 8:22pm			100 / 100
Lab - 2 Build your VPC and Launch a Web Server Labs	Oct 11 at 7:43pm			100 / 100
Lab - 3 Introduction to Amazon EC2 Labs				- / 100

- Assuming you have an AWS VPC with a public and private subnet, describe how you would configure a NAT Gateway to provide internet connectivity to EC2 instances running in the private subnet without those instances needing to have a direct connection to the internet gateway.

Ans 2) Setting up a NAT Gateway to connect a private subnet.

Comprehending the Situation:

Private subnet: Instances within this subnet are unable to communicate directly with the internet.

NAT Gateway: An apparatus for network address translation enabling private IP addresses to access the internet via a public IP address.

An Internet Gateway is a connection point from your VPC to the internet.

Steps to set up/configure:

Set up a NAT Gateway:

Select a public subnet: The NAT gateway needs to be located in a public subnet in order to connect to the internet.

Choose a public IP address: Allocate a public IP address to the NAT gateway.

Set up the bandwidth: Select the suitable bandwidth for your requirements.

Set up a Route Table:

MGS 602: Network Technology and Cloud Infrastructure Management

Connect the private subnet to the route table to route traffic through the NAT gateway.

Establish a pathway:

Destination: 0.0.0.0/0 (default route) - no specific destination address specified

Goal: The identification of the NAT gateway

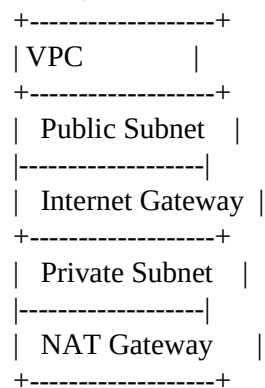
Change the Route Table of the Internet Gateway.

Locate the standard route: The target is 0.0.0.0/0.

Amend the goal: Update the objective to the ID of the NAT gateway. This guarantees that the NAT gateway routes all internet traffic from the VPC.

3. A diagram summarizing the VPC resources that have been created, including the VPC, subnets, route tables, and network connections.

Ans 3) Here's the diagram:



4. A brief explanation of the purpose and function of an internet gateway and a NAT gateway, and how they enable communication between EC2 instances in a VPC and the internet.

Ans 4) Internet Gateway (IGW) is a device that allows communication between devices within a network and the internet.

Function: serves as a connection between a VPC and the internet.

Purpose: Allows for direct communication between EC2 instances in the VPC and the internet.

How it operates: Creates a pathway for internet traffic.

Network Address Translation Gateway

Objective: Offers translation of network addresses for EC2 instances located in a private subnet.

Purpose: Permits EC2 instances located in a private subnet to reach the internet without revealing their private IP addresses publicly.

MGS 602: Network Technology and Cloud Infrastructure Management

The process involves changing private IP addresses to public IP addresses to allow communication with the internet via the IGW.

As a team:

VPC includes both public and private subnets: EC2 instances in the public subnet have direct internet access through the IGW.

NAT Gateway for private subnet: The NAT gateway allows EC2 instances in the private subnet to connect to the internet by converting their private IP addresses to public ones.

Connection: Both kinds of instances are able to connect to the internet, guaranteeing connectivity while upholding security for instances located in the private subnet.